



Direct Numerical Simulations of Turbulent Channel and Duct Flows Using Interpolation-Based Lattice Boltzmann Method

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Received: 20 February 2025 / Accepted: 18 August 2025 / Published online: 17 September 2025
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Abstract

The Lattice Boltzmann Method (LBM), a weakly compressible approach, has attracted considerable attention for turbulent flow simulations but remains limited in high-Reynolds-number applications due to its reliance on uniform meshes. This study investigates the interpolation-based Lattice Boltzmann Method (IBLBM) for direct numerical simulation (DNS) of turbulent channel and duct flows using nonuniform grids. Simulations are conducted for turbulent Poiseuille channel flows at $Re_\tau = 180, 395, 640$, and **1000**, as well as for turbulent duct flow at $Re_\tau = 360$. The predicted mean velocity profiles, Reynolds stresses, vorticity distributions, and turbulent kinetic energy budgets show excellent agreement with benchmark DNS data. The IBLBM scheme accurately resolves near-wall turbulence and secondary flow structures in ducts while maintaining similar computational overhead across different interpolation orders. For the 7- and 200-million-grid cases at $Re_\tau = 180$ and 640, LBM with $\Delta^+ \sim 3$ is approximately 1.9 and 3.7 times faster, respectively, than sixth-order IBLBM; however, to match IBLBM's wall resolution ($\Delta^+ \sim 0.3$) using $\Delta^+ \sim 1$, LBM becomes four to six times slower. These results demonstrate the robustness and accuracy of IBLBM for high-Reynolds-number wall-bounded turbulence, offering an optimal balance between computational efficiency and physical fidelity through the use of nonuniform grids.

Keywords Lattice boltzmann method · Direct numerical simulation · Interpolation · Non-uniform mesh · Turbulent channel flow · Turbulent duct flow · Graphics processing unit

1 Introduction

The Lattice Boltzmann method (LBM) (Aidun and Clausen 2010) has gained prominence as a noteworthy alternative to Navier-Stokes solvers over the past three decades. The explicit formulation of LBM makes it particularly well suited for parallel computations, and its ease of implementation further enhances its appeal. These characteristics make LBM highly efficient in addressing a wide range of flow problems, such as pressure-driven flows (Lin et

al. 2020; Wang et al. 2024), flows involving natural convection (Dubois et al. 2016), two-phase flows (Huang et al. 2018), and porous media flows (Kuwata et al. 2020). Despite its advantages, the original LBM is limited by its dependence on uniform Cartesian meshes, which poses a notable constraint when simulating complex flows. This restriction becomes particularly evident in wall-bounded turbulent flows, where the accurate resolution of the near-wall region necessitates the use of fine meshes (Hsu et al. 2012).

To overcome the constraints imposed by the necessity of uniform meshes, various investigations have used multiblock grid refinement techniques (Filippova and Hänel 1998; Wu et al. 2020). These methodologies facilitate adaptable adjustment of grid resolution to cater to the specific requirements of the flow field. Despite yielding favorable outcomes, these approaches maintain the use of uniform meshes within individual blocks. As a consequence, the handling of interfaces between distinct blocks remains a nontrivial challenge.

Various alternative approaches have been explored to mitigate the limitations associated with uniform mesh requirements. One such strategy involves the adoption of nonuniform meshes, where the coupling between the discretization of momentum and physical space is relaxed (He et al. 1996; He and Doolen 1997). Despite the localization of collisions, the differential mesh width impedes the streaming operation from reaching adjacent grid nodes. Consequently, interpolation techniques are used to recover the distribution function within the computational mesh. The selection of an interpolation stencil significantly influences the accuracy of predictions, with He et al. (1996) demonstrating that quadratic interpolation achieves better accuracy, albeit in simulations conducted at low Reynolds laminar numbers.

Furthermore, Cheng and Hung (2004) employed an upward-based Lagrangian interpolation technique, whereas Krämer et al. (2017) introduced the concept of the Semi-Lagrangian Lattice Boltzmann Method (SLLBM). In SLLBM, the precision of the reconstruction of the distribution function is enhanced through the integration of the finite element method, albeit at the cost of requiring a substantial amount of computational memory. In particular, Di Ilio et al. (2018) conducted simulations of turbulent flow past a circular cylinder at a Reynolds number of $Re=3900$ using second-order SLLBM on body-fit meshes. These nonuniform mesh approaches offer promising solutions to simulate complex flows, providing greater flexibility and accuracy compared to traditional uniform mesh methodologies.

The aforementioned methodologies are mainly based on the Lagrangian formulation. An alternative avenue involves employing the Eulerian form of the discrete-lattice Boltzmann equation (LBE). In this paradigm, the LBE can be addressed analogously to the convection equation, utilizing the finite volume method (Nannelli and Succi 1992; Peng et al. 1998; 1999) or the finite difference method (Lee and Lin 2003).

In this Eulerian approach, the collision term in the LBE is treated as a source term, and the precision of both the time derivative and the advection term discretizations directly influences the temporal and spatial accuracy of the solution. Additionally, the finite-volume or finite-difference methods are employed to discretize the governing equations, facilitating fluid flow simulation from an Eulerian perspective. These techniques offer an alternative route for lattice Boltzmann simulations, presenting distinct trade-offs in terms of accuracy, stability, and computational efficiency.

Pressure-driven turbulent channel flows (Moser et al. 1999; Abe et al. 2001; Lee and Moser 2015) have served as benchmark cases to assess the validity of direct numerical simulation studies of the Lattice Boltzmann method (LBM-DNS) (Suga et al. 2015; Gehrke et al. 2017; Lee et al. 2018). The predominant focus of existing LBM-DNS investigations lies

in the realm of low Reynolds numbers, typically around $Re_\tau \sim 180$ ($Re_{bulk} \sim 5660$) (Suga et al. 2015; Lee et al. 2018).

Nathen et al. (2018) conducted LBM-DNS at lower resolutions for channel flows at $Re_\tau = 590$. The uniform grid resolution is approximately $\Delta^+ \sim 6.48$, leading to a 20% under-prediction of stresses due to this low resolution. In particular, the highest Reynolds number achieved in LBM-DNS is $Re_\tau = 640$ ($Re_{bulk} = 24000$) (Wu et al. 2020), achieved through successive block mesh refinement strategies with a grid count of 235×10^6 using $\Delta_{min}^+ \sim 1$, albeit with minor discontinuities at the block mesh interface. Recently, Xiang et al. (2023) conducted turbulent duct flow up to $Re_\tau = 2000$ based on duct height using D3Q19 SRT LBM and uniform mesh with a grid resolution of approximately $\Delta^+ \sim 3.4$. However, Lee et al. (2018) and Kand and Hassan (2013) indicated that the D3Q19 lattice model is unsuitable for turbulent duct flow simulations, as the model generates additional corner vortex. The anisotropy and associated defects of the D3Q19 model have also been observed in jet flow simulations, as noted by Geller et al. (2013).

As indicated by Smits and Marusic (2013), industrial applications demand Reynolds numbers exceeding $Re_\tau \gtrsim 1000$ ($Re_{bulk} \sim 40,000$), thereby necessitating a thorough investigation of the proposed scheme. Accordingly, this study employs direct numerical simulation (DNS) of channel flows up to $Re_\tau \sim 1000$ to evaluate the method's capability and performance under industrially relevant conditions. The selected grid density aligns with that used in Navier–Stokes equation-based simulations (Moser et al. 1999; Abe et al. 2001; Lee and Moser 2015), enabling a comparative assessment of the interpolation-based lattice Boltzmann method (IBLBM) across different interpolation orders.

To date, performance evaluations beyond second-order interpolation remain scarce, particularly in the context of complex turbulent flows. The present interpolation strategies address challenges arising from nonuniform mesh widths, which inhibit the streaming process from naturally reaching adjacent grid nodes. Furthermore, the method readily accommodates different collision models, such as the single-relaxation-time and multi-relaxation-time lattice Boltzmann methods (LBM) (Aidun and Clausen 2010), due to the inherently localized nature of the collision operation.

The paper is organized into distinct sections to facilitate a comprehensive understanding of the presented research. In Sect. 2, the formulation of the lattice Boltzmann method is explained, accompanied by a detailed discussion of the interpolation strategies used. Section 3 offers an extensive exposition on the results derived from the analysis of the laminar Poiseuille flow and the turbulent Poiseuille flow. Finally, Sect. 4 succinctly summarizes the key findings and conclusions drawn from the study.

2 Methodology

2.1 Lattice Boltzmann Method

The evolution of the distribution function of the single relaxation time (SRT) LBM reads (Aidun and Clausen 2010),

$$f_i(\mathbf{x} + \mathbf{e}_i \delta t, t) = f_i(\mathbf{x}, t - \delta t) - \frac{1}{\tau} [f_i(\mathbf{x}, t - \delta t) - f_i^{eq}(\mathbf{x}, t - \delta t)] + F_i \delta t \quad (1)$$

where τ , f_i , and f_i^{eq} are the relaxation time, the distribution function, and the equilibrium distribution, respectively. F_i is the body force, and the model proposed by He et al. (1997) is used, that is,

$$F_i = \omega_i \frac{\rho}{c_s^2} \mathbf{e}_i \cdot \mathbf{G}. \quad (2)$$

The equilibrium distribution function is given as

$$f_i^{eq} = \omega_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{1}{2} \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{c_s^4} - \frac{1}{2} \frac{\mathbf{u} \cdot \mathbf{u}}{c_s^2} \right], \quad (3)$$

where ω_i is the weighting coefficients corresponding to the discrete velocity, $\mathbf{e}_i$. The speed of sound is given by $c_s = c/\sqrt{3}$, where the lattice velocity is defined as $c = \delta x/\delta t = 1$, assuming that the lattice spacing δx equals the lattice time step δt . ρ and $\mathbf{u}$ are macro density and velocity, respectively, which can be computed as

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i \mathbf{e}_i f_i. \quad (4)$$

To simplify the numerical procedure, we split Eq. (1) into two steps.

(1) Collision:

$$f_i^+ (\mathbf{x}, t) = f (\mathbf{x}, t) - \frac{1}{\tau} [f_i (\mathbf{x}, t) - f_i^{eq} (\mathbf{x}, t)] + F_i \delta t \quad (5)$$

(2) Streaming:

$$f_i (\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) = f_i^+ (\mathbf{x}, t) \quad (6)$$

This work uses the D2Q9 and D3Q19 and D3Q27 models to simulate the 2-D and 3-D flows. The discrete velocities in the D2Q9 model are given as $e_0 = (0, 0)$, $e_1 = -e_3 = (1, 0)$, $e_2 = -e_4 = (0, 1)$, $e_5 = -e_7 = (1, 1)$, and $e_6 = -e_8 = (-1, 1)$. The weighting coefficients are $\omega_0 = 4/9$, $\omega_{1-4} = 1/9$, and $\omega_{5-8} = 1/36$, respectively. In the D3Q19 and D3Q27 models,

$$e_i = \begin{cases} (0, 0, 0), & i = 0, \\ (\pm 1, 0, 0), & (0, \pm 1, 0), & (0, 0, \pm 1), & i = 1 \sim 6, \\ (\pm 1, \pm 1, 0), & (\pm 1, 0, \pm 1), & (0, \pm 1, \pm 1), & i = 7 \sim 18, \\ (\pm 1, \pm 1, \pm 1), & & & i = 19 \sim 26, \end{cases} \quad (7)$$

and D3Q19: $\omega_0 = 1/3$, $\omega_{1-6} = 1/18$, $\omega_{7-18} = 1/36$, D3Q27: $\omega_0 = 8/27$, $\omega_{1-6} = 2/27$, $\omega_{7-18} = 1/54$, $\omega_{19-26} = 1/216$.

The adaptation to other collision operators, such as the multi-relaxation-time (MRT) lattice Boltzmann method described by Lin et al. (2020), can be readily implemented due to the inherently localized nature of the collision process. In the present study, preliminary

investigations indicate that, under the current nonuniform mesh configuration, the differences between MRT and single-relaxation-time (SRT) models are marginal. Therefore, the SRT model is adopted for the simulations. In the following section, we introduce interpolation strategies designed to address challenges posed by differential mesh widths, which impede the streaming process from reaching neighboring grid nodes.

2.2 The Interpolation-Based Lattice Boltzmann Method

During the streaming step, meticulous consideration must be given to the departure lattice point, $\mathbf{x}_{dep}$, at $\mathbf{x} - \mathbf{e}_i \delta t$ due to potential misalignment with the computational mesh point and lattice point, particularly in nonuniform grids, i.e., $\Delta x, \Delta y, \Delta z \neq \delta x$. To rectify this misalignment, the present study employs an interpolation process directly within the physical domain. This interpolation process is instrumental in determining the values of the post-collision distribution function, denoted $f^+(\mathbf{x}_{dep})$, at the departure points. The rationale behind adopting this approach is grounded in the assumption that, in the context of incompressible flow, physical properties exhibit continuity throughout the entire physical domain, as postulated by He et al. (1996).

The value of the post-collision distribution at the departure point is the summation of the value of each support point multiplied by its respective interpolation coefficients. The equation can be expressed as follows:

$$f^+(\mathbf{x}_{dep}) = \sum_{s=0}^{n^d} L_s f^+(\mathbf{x}_s). \quad (8)$$

Equation 8 is a versatile formula that can be applied to various interpolation orders (IOs), represented by the variable n , and to physical problems of different dimensions, spanning from one to three, denoted by $d=1, 2$, and 3 , respectively. The shape function $L_{n,s}$ corresponds to the Lagrange polynomial associated with $f^+(\mathbf{x}_s)$ and is evaluated at the respective support points.

The arrangement of support points on a two-dimensional (2D) nonuniform grid is illustrated in Fig. 1(a), using a fourth-order interpolation as an example. The red open circles indicate the departure points ($\mathbf{x}_{dep}$), while the support points ($\mathbf{x}_s$) are shown as red filled circles. The lattice width (δx), outlined by the dashed square in Fig. 1, may differ from the physical mesh size (Δx or Δy), as evidenced by the spacing between the filled circles in each direction.

These support points are symmetrically distributed around the target point (enclosed by the black dashed square), as illustrated in Fig. 1(a). To implement boundary conditions, the halfway bounce-back scheme (Lee et al. 2018; Ziegler 1993) is employed, which inherently lacks nodes beyond the domain boundary. Consequently, for grid points near the wall—as shown in Fig. 1(a)—a biased stencil is adopted due to the absence of support points outside the wall. This approach ensures that the interpolation process maintains a consistent level of accuracy across all points, including those adjacent to the boundary.

Figure 1(b) provides a zoomed-in view of an interpolation stencil, offering a detailed look at the interpolation process. For example, in the interpolation of the two-dimensional flow field, as shown in Fig. 1(a), the interpolation coefficients for each support point encom-

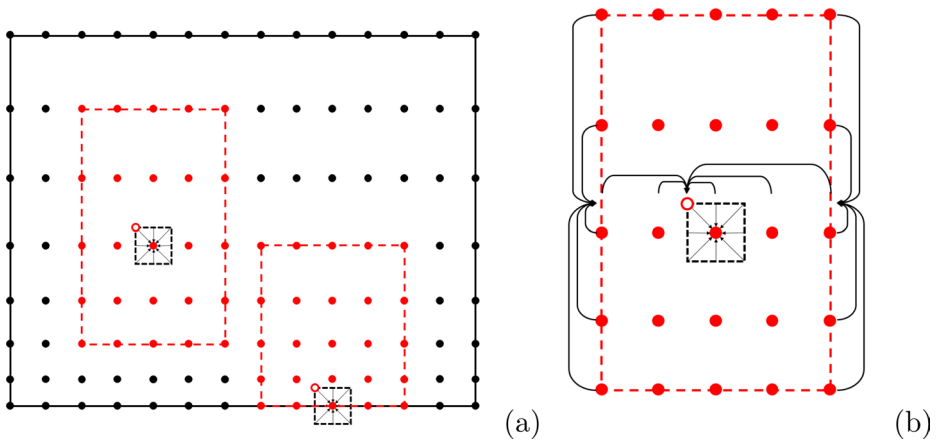


Fig. 1 Support points of ILBM; filled red circle: support points $\phi(\mathbf{x}_s)$; open red circle: departure point $\phi(\mathbf{x}_{dep})$. **(a)** illustrates the distribution of support points on a 2D nonuniform grid, with the target point located at the center of the domain and biased stencils applied near boundaries to compensate for missing nodes. **(b)** shows an enlarged view of the support stencil, providing a detailed look at the Lagrangian interpolation process

pass two components: one in the x direction and the other in the y direction. In this scenario, the Lagrange polynomial, denoted as L_s , can be expressed as

$$L_s(\mathbf{x}_s) = \prod_{0 \leq i \leq n, i \neq s} \frac{\mathbf{x}_{dep} - \mathbf{x}_i}{\mathbf{x}_s - \mathbf{x}_i} \prod_{0 \leq j \leq n, j \neq s} \frac{\mathbf{y}_{dep} - \mathbf{y}_j}{\mathbf{y}_s - \mathbf{y}_j} \quad (9)$$

As shown in Eq. 9, the shape functions L_s used in this study depend solely on the positions of the support points and the departure points. Their values remain constant throughout the simulation. To reduce computational overhead, the values of each L_s are precomputed and stored prior to the simulation, following the strategy employed in previous studies on the semi-Lagrangian lattice Boltzmann method (SLLBM) (Di Ilio et al. 2018; Krämer et al. 2020).

2.3 GPU Implementation

In this investigation, 3-dimensional (3D) flow-field simulations are performed on multi-GPU clusters. The GPU implementation builds upon previous research efforts (Lee et al. 2018; Chang et al. 2013; Hong et al. 2015). To efficiently manage the computational workload, a one-dimensional domain decomposition strategy is employed, as illustrated in Fig. 2.

Each subdomain is assigned to a dedicated GPU, and the decomposition is performed along the streamwise direction (e.g., the x -axis), dividing the global domain into equally sized slabs, shown as green and dark blue boxes in Fig. 2. Within each GPU's memory, the allocated storage for distribution functions is slightly enlarged to include buffer regions that support interpolation near subdomain boundaries, depicted as light blue boxes in Fig. 2. The number of buffer layers depends on the interpolation order: fourth-order interpolation requires two layers on each side of a subdomain, whereas sixth-order interpolation requires three.

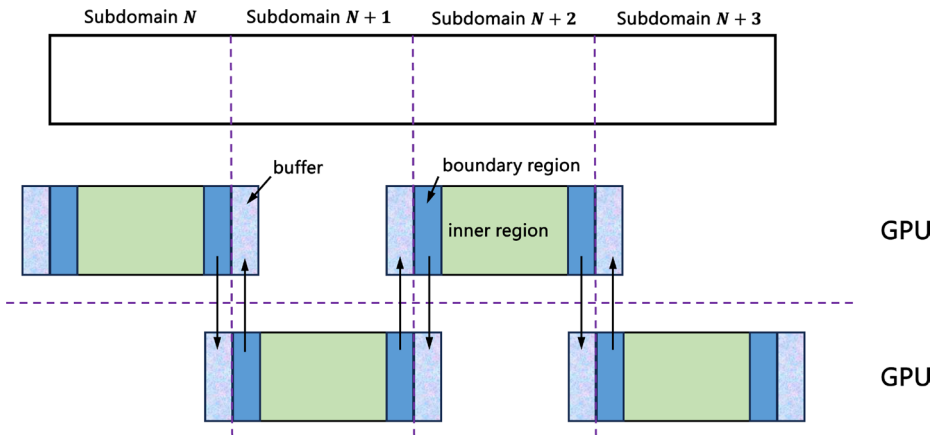


Fig. 2 One-dimensional domain decomposition and communication pattern. Each subdomain is assigned to a GPU and divided into inner and boundary regions, green and dark blue boxes, respectively. Buffer layers are used to exchange boundary data between neighboring GPUs

Communication between neighboring GPUs is facilitated via GPUDirect Remote Direct Memory Access (RDMA) (2025), which enables accelerated data transfers between GPUs, especially when used in conjunction with CUDA-aware MPI for inter-node communication. At each time step, the streaming distribution functions near the subdomain boundaries—specifically, the 10 directions out of 19 in the D3Q19 lattice model that are relevant to the one-dimensional domain decomposition—are exchanged between adjacent GPUs.

To improve scalability and computational performance, the implementation overlaps communication with computation, following the approach adopted by Hong et al. (2015), effectively hiding the latency associated with data transfers. While data exchange occurs at subdomain boundaries, interior grid points that do not require updated halo data are computed concurrently.

3 Results

3.1 Simulations in Fully Developed Laminar Poiseuille Channel Flow with Wall Injection and Suction

The simulations are directed to the Poiseuille flow with non-permeable walls, that is, ($V = V_w$). The analytical solution reads

$$u = \frac{h^2}{Re_w} \frac{G_x}{\nu} \left[\frac{y-h}{h} - 1 + \frac{e^{Re_w} - e^{Re_w(y-h)/h}}{\sinh(Re_w)} \right] \quad (10)$$

where $Re_w = V_w h / \nu$ and $H = 2h$ is the channel height. Without the permeable wall, i.e., $V_w = 0$, the streamwise pressure gradient, G_x , determines the maximum velocity in the channel, given by U_{max} . Consequently, the Reynolds number in the absence of a permeable wall can be defined as $Re_o = U_{max} H / \nu$.

The nonuniform physical mesh can be generated using a hyperbolic tangent function (Abe et al. 2001; Shi and Lin 2020; Shi et al. 2020), given by:

$$y_j = \frac{H}{a} \tanh \left(\frac{-1 + 2j/N_y}{2} \ln \frac{1+a}{1-a} \right). \quad (11)$$

Here, N_y represents the number of grid points in the vertical direction, and $a \in (0, 1)$ is an adjustable parameter. In this injection and suction problem, the velocity ratio is set to $a=0.9$. The lattice spacing is chosen based on the minimum physical mesh size, i.e., $\delta x = \Delta y_{\min}$. Four sets of grid resolutions are considered: $N_y = 48, 64, 128$, and 256 . The corresponding grid sizes used in both LBM and IBLBM are summarized in Table 1. As shown, the uniform grid spacing (Δy) in LBM is approximately three times larger than the minimum lattice spacing employed (Δy) in IBLBM when $a=0.9$.

In these simulations, the Reynolds numbers $Re_o = 30$ and $Re_w = 10$ are used. By incorporating a vertical velocity, the maximum velocity is anticipated to shift towards the upper boundary, resulting in a more pronounced velocity distribution at the top wall. This behavior resembles that observed in turbulent boundary layers, where the nonuniform mesh offers distinct advantages. Additionally, as velocity distributions deviate from quadratic profiles, these cases serve as test scenarios to analyze the effects of interpolation.

Additionally, complementary Navier–Stokes solutions computed on the same non-uniform mesh are included for reference. These results are obtained by solving the following discretized form of the Navier–Stokes Eq. (12):

$$\frac{U_{j+1}V_{j+1} - U_{j-1}V_{j-1}}{y_{j+1} - y_{j-1}} = 2\nu \frac{\frac{U_{j+1}-U_j}{y_{j+1}-y_j} - \frac{U_j-U_{j-1}}{y_j-y_{j-1}}}{y_{j+1} - y_{j-1}} + G_x \quad (12)$$

where G_x represents the pressure gradient for fully developed flow at Reynolds number Re_o .

Figure 3 presents the velocity distribution predicted by the IBLBM, revealing an upward shift in the peak of the streamwise velocity. This shift highlights the inherent advantage of using a non-uniform mesh over a uniform one. This benefit becomes more pronounced when examining the L_∞ norm distribution with varying numbers of grid points (N_y), as shown in Fig. 4. The figure compares IBLBM simulations with different interpolation orders on non-uniform meshes against standard LBM results using uniform meshes.

The second-order accuracy of the lattice Boltzmann method is prominently demonstrated in Fig. 4, although with exceptions noted for second-order interpolation at lower grid resolutions. Specifically, at equivalent grid resolutions, the interpolation orders of 4th and 6th exhibit comparable distributions, indicating superior accuracy.

Table 1 Grid distribution of the flow with transpiration

N_y	LBM	IBLBM		
	Δy	$\Delta y_{\min}$	$\Delta y_{\max}$	δx
48	H/48	$0.329\Delta y$	$1.634\Delta y$	$0.329\Delta y$
64	H/64	$0.324\Delta y$	$1.635\Delta y$	$0.324\Delta y$
128	H/128	$0.317\Delta y$	$1.636\Delta y$	$0.317\Delta y$
256	H/256	$0.314\Delta y$	$1.636\Delta y$	$0.314\Delta y$

Fig. 3 Predicted velocity distribution with permeable wall using the IBLBM ($Re_o = 30$, $Re_w = 10$.)

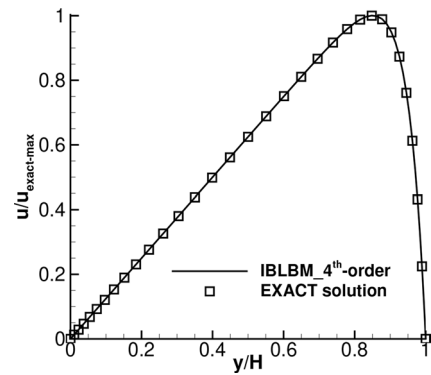


Fig. 4 L_∞ norm at different number of mesh ($Re_o = 30$, $Re_w = 10$.)

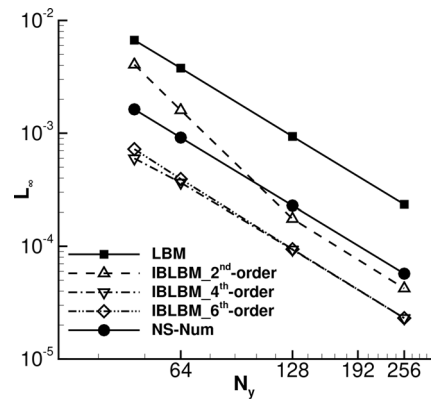


Fig. 5 L_∞ norm at different lattice spacing δ/H . ($Re_o = 30$, $Re_w = 10$.)

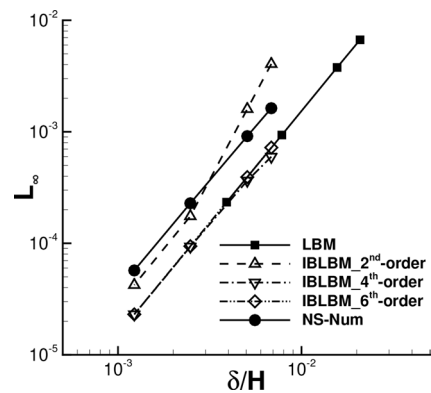


Figure 5 illustrates the error norms of Fig. 4 with respect to the lattice spacing (δy) employed. LBM and 4th order and 6th order IBLBM consistently depict similar distributions, while 2nd order ISLBM shows decreased accuracy, particularly under this complex flow condition. Thus, the present ISLBM at high interpolation order, in particular, 4th order and 6th order, can produce the same performance as regular LBM results but at a lower grid

density. For instance, the maximum error of the LBM results with $N_y = 256$ is comparable to that obtained using $N_y = 96$ for the 4th and 6th-order IBLBM, as shown in Fig. 4.

3.2 Turbulent Poiseuille Channel Flows

The current numerical procedure is used to calculate turbulent channel flows bounded by walls driven by pressure. Non-slip boundary conditions are applied in the y direction, while periodic conditions are enforced in the x and z directions. A constant pressure driving force is included in the streamwise direction to attain the target Reynolds numbers.

In this particular setup of wall-bounded flow, grid refinement is intentionally focused near the walls to resolve the thin viscous sublayer adequately. As a result, a nonuniform grid is generated in the vertical direction utilizing a hyperbolic tangent function, as elaborated in Eq. 11.

Four Reynolds numbers are under investigation: $Re_\tau = 180, 395, 640$, and 1000 , based on the friction velocity and the half-height of the channel. The $Re_\tau = 180$ case has been extensively studied, while $Re_\tau = 1000$ has not been explored using direct numerical simulations with the lattice Boltzmann method. Table 2 provides a comprehensive overview of the grid configurations and lattice dimensions used throughout the study.

To ensure efficient acquisition of results through direct numerical simulation while maintaining a reasonable mesh size, subsequent scenarios maintain a grid spacing of approximately $\Delta^+ \sim 5$ in both the span and the streamwise directions. Furthermore, the vertical resolution is kept within the approximate range of 0.3 to 5 (Abe et al. 2001). These grid densities were considered sufficient for direct numerical simulation using the Navier-Stokes equation.

Figure 6 depicts the predicted streamwise velocity utilizing various interpolation orders (IO: 2 to 8) at Reynolds numbers $Re_\tau = 180$ (IO: 2,4,6), 640 (IO: 6,8), and 1000 (IO: 6,8). At $Re_\tau = 180$, the 2nd order interpolation presents an excessively accelerated profile, whereas the 4th and 6th order interpolations exhibit similar results, though the 6th order results are marginally better. The impacts of these interpolation orders are further scrutinized by contrasting the turbulent vorticity distributions, as shown in Fig. 7. Similarly to observations in streamwise velocity distributions at $Re_\tau = 180$, the 2nd order interpolation deviates from the benchmark solutions. In contrast, the 4th and 6th order interpolations show similar results, with the latter being marginally better.

At a higher Reynolds number of $Re_\tau = 640$, we investigated the effects of interpolations of the order 6th and 8th, as shown in Figs. 6 and 7, which reveal marginally different results. This is also true for the $Re_\tau = 1000$ case, where marginal difference results are observed. Consequently, for subsequent simulations, we adopt 6th-order interpolation for the Reynolds number investigated.

The performance of the current interpolation-based lattice Boltzmann method (IBLBM) in simulating mean velocity profiles across various Reynolds numbers up to $Re_\tau = 1000$ is evident from Fig. 8, which demonstrates excellent alignment with benchmark solutions using direct numerical simulation data of Moser et al. (1999); Abe et al. (2001, 2004), and Lee and Moser (2015). In particular, the model accurately captures the heightened mean velocities beyond the logarithmic region at high Reynolds numbers in the center of the channel.

Table 2 Grid setting for the turbulent channel flow at $Re_\tau = 180, 395, 640$, and 1000

Re_τ	Domain size	Grid number	Δx^+	Δz^+	Δy^+	y_w^+	lattice size δ^+
180	$6.4h \times 2h \times 3.2h$	$192 \times 192 \times 192$	6.00	3.00	$0.30 \sim 3.83$	0.15	0.30
395	$6.4h \times 2h \times 3.2h$	$512 \times 320 \times 384$	4.89	3.26	$0.28 \sim 5.40$	0.14	0.28
640	$6.4h \times 2h \times 3.2h$	$832 \times 512 \times 464$	4.92	5.33	$0.31 \sim 5.47$	0.15	0.31
1000	$6.4h \times 2h \times 3.2h$	$1232 \times 768 \times 608$	5.19	5.26	$0.41 \sim 5.26$	0.20	0.41

Further insight into the predicted Reynolds stress is provided in Fig. 9, which shows the root mean square of normal stresses and shear stress. The model successfully anticipates the variations of the turbulence levels corresponding to changes in the Reynolds numbers, displaying strong agreement with established benchmark solutions (Moser et al. 1999; Abe et al. 2001; Lee and Moser 2015; Abe et al. 2004).

The distributions of the variation of the root mean square (rms) vorticity are illustrated in Fig. 10, demonstrating the correctly predicted increase in vorticity along the stream and span directions as the Reynolds number increases. These predicted results exhibit a favorable comparison with established benchmark solutions (Moser et al. 1999; Abe et al. 2001; Lee and Moser 2015).

The turbulence transport mechanism can be best exemplified by the turbulent kinetic energy equation (Hsu et al. 2012; Hwang and Lin 1998), which can be expressed as

$$-\underbrace{\frac{\partial}{\partial x_j^+} \overline{u_j^+ k'^+}}_{C^+} - \underbrace{\overline{u_i'^+ u_j'^+} \frac{\partial \overline{u_i^+}}{\partial x_j^+}}_{P^+} - \underbrace{\nu \frac{\partial^2 \overline{k'^+}}{\partial x_j^+ \partial x_j^+}}_{d^{+v}} - \underbrace{\frac{\partial \overline{u_j'^+ k'^+}}{\partial x_j^+}}_{d^{+t}} - \underbrace{\frac{\partial \overline{u_j'^+ p'^+}}{\partial x_j^+}}_{d^{+p}} - \underbrace{\nu \frac{\partial \overline{u_i'^+}}{\partial x_j^+} \frac{\partial \overline{u_i^+}}{\partial x_j^+}}_{\epsilon^+} = 0$$

where $k'^+ = \overline{u_i'^+ u_i'^+} / 2$ is the turbulent kinetic energy and C^+ : convection, P^+ : production, d^{+v} : viscous diffusion, d^{+p} : pressure diffusion, ϵ^+ : turbulent dissipation rate. For the present fully developed flow, the convection term is zero.

Figure 11 presents the turbulent kinetic energy budget at Re_τ values of 180, 395, 640 and 1000, along with reference solutions from Moser et al. (1999), Abe et al. (2001, 2004), and Lee and Moser (2015). The predicted results of the current IBLBM align favorably with these benchmark solutions. The model accurately captures the increase in the viscous diffusion in the near-wall and the turbulence dissipation rate in response to escalating Reynolds numbers. Furthermore, it correctly predicts the plateau region of the turbulence dissipation rate around $y^+ \sim 10$.

Furthermore, we evaluate the GPU performance of the interpolation-based Lattice Boltzmann Method (IBLBM) using different interpolation orders at friction Reynolds numbers $Re_\tau = 180$ and 640, employing four and eight NVIDIA Tesla V100 GPUs, respectively. The performance metric is the million lattice updates per second (MLUPS), defined as:

$$\text{MLUPS} = \frac{\text{grid size} \times \text{number of iterations} \times 10^{-6}}{\text{total simulation time}}. \quad (13)$$

The simulation time is measured over 10,000 time steps.

The results are summarized in Tables 3 and 4, corresponding to simulations with approximately 7 million and 200 million grid points, respectively, for LBM and IBLBM under

Fig. 6 Predicted streamwise velocity distributions using different interpolation orders at $Re_\tau = 180, 640$ and 1000. (dash-dot: 2nd-order, dashed: 4th-order, solid: 6th-order and dash-do-dot: 8th-order)

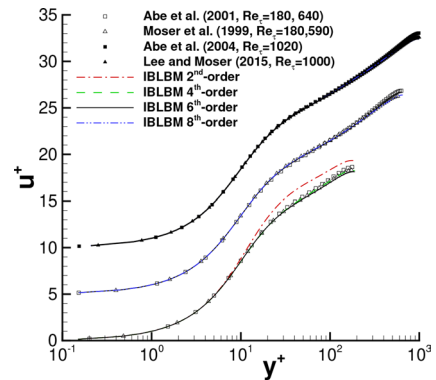


Fig. 7 Predicted turbulent vorticities distributions using different interpolation orders at $Re_\tau = 180$ and 640. (dash-dot: 2nd-order, dashed: 4th-order, solid: 6th-order and dash-do-dot: 8th-order); $\square$: Moser et al. (1999), Δ : Abe et al. (2001)

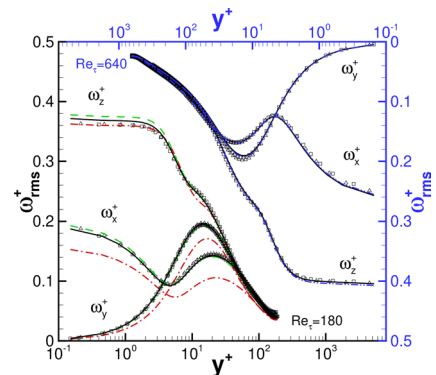
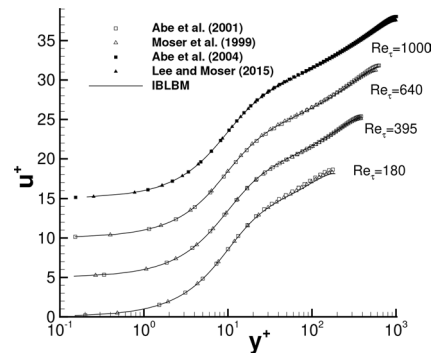


Fig. 8 Predicted streamwise velocity distributions at different reynolds numbers. ($Re_\tau = 180, 395, 640$ and 1000)



varying interpolation orders. Despite employing a similar total number of grid points, the effective resolution differs. For the standard LBM grid, the near-wall resolution is $\Delta^+ \sim 3-4$, which is considered coarse. Achieving a finer resolution of $\Delta^+ \sim 1$ with LBM would require 37 to 64 times more grid points than the current IBLBM configurations.

For the 7-million-grid case, LBM achieves performance that is approximately 90% higher than IBLBM with sixth-order interpolation. For the 200-million-grid simulation at $Re_\tau = 640$, LBM demonstrates a performance improvement of about 3.7 times over the

Fig. 9 Predicted turbulence intensity distributions at different reynolds numbers. ($Re_\tau = 180, 395, 640$ and 1000)

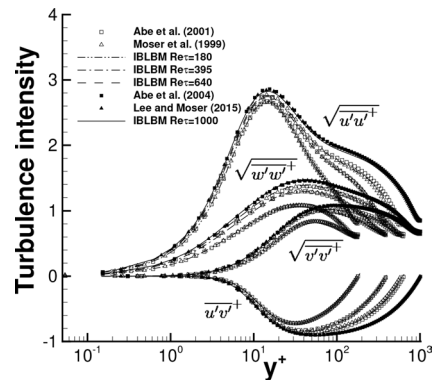


Fig. 10 Predicted turbulent vorticity distributions at different reynolds numbers. ($Re_\tau = 180, 395, 640$ and 1000)

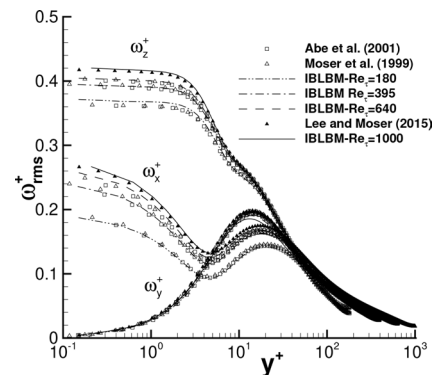
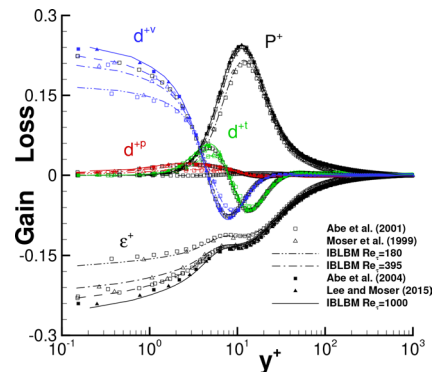


Fig. 11 Predicted turbulent kinetic energy budgets at different reynolds numbers. ($Re_\tau = 180, 395, 640$ and 1000)



sixth-order IBLBM. However, if the LBM were to match the wall resolution of IBLBM ($\Delta^+ \sim 1$), the associated grid size and computational cost would increase dramatically, diminishing the advantage.

Interestingly, the MLUPS per GPU for IBLBM using second- and fourth-order interpolation remains relatively stable across different grid sizes. In contrast, the performance of conventional LBM improves significantly, increasing from 895 MLUPS to 1202 MLUPS per GPU as the grid size grows. On the other hand, IBLBM with sixth-order interpolation

Table 3 GPU performance of $Re_\tau = 180$ channel flow on four NVIDIA V100 GPUs for LBM (with grid number $360 \times 112 \times 180$) and IBLBM (with grid number $192 \times 192 \times 192$) in different order interpolations

$Re_\tau = 180$ grid size resolution	LBM 7.26×10^6 $\Delta_{x,y,z}^+ = 3.2$	IBLBM 7.07×10^6 $\Delta^+ = 6, 0.30 \sim 3.83, 3$			
	2nd-order		4th-order	6th-order	
time (s)	20.3	21.4	32	37.8	
MLUPS	3578	3301	2211	1873	
MLUPS/GPU no.	895	825	553	468	
MLUPS _{LBM} /MLUPS	1	1.09	1.62	1.91	

Table 4 GPU performance of $Re_\tau = 640$ channel flow on eight NVIDIA V100 GPUs for LBM (with grid number $1088 \times 340 \times 544$) and IBLBM (with grid number $832 \times 512 \times 464$) in different order interpolations

$Re_\tau = 640$ grid size resolution	LBM 201×10^6 $\Delta^+_{x,y,z} = 3.8$	IBLBM 198×10^6 $\Delta^+ = 4.92, 0.31 \sim 5.47, 4.41$		
		2nd-order	4th-order	6th-order
time (s)	209	296	452	753
MLUPS	961.5	6660	4369	2623
MLUPS/GPU no.	1202	833	546	328
MLUPS _{LBM} /MLUPS	1	1.44	2.07	3.67

exhibits a notable decline in performance, decreasing from 468 MLUPS to 328 MLUPS per GPU. The underlying cause of this degradation is not yet fully understood, but it may be attributed to GPU occupancy limitations, possibly due to the partitioned matrix size and buffer memory usage in LBM and IBLBM implementations.

The computational time required per eddy turnover time is also included, as illustrated in Tables 5 and 6, which is used to account for the reduction of the time step if smaller lattice unit is used. It should be noted that the CPU time for the refined-grid LBM cases is estimated based on the increase in grid density under the assumption of perfect scaling.

As shown in Table 5 for the $Re_\tau = 180$ case, LBM with a grid resolution of $\Delta^+ = 1.07$ requires nearly four times more CPU time per eddy turnover than IBLBM with sixth-order interpolation, even though the near-wall grid resolution of IBLBM ($\Delta^+ = 0.30$) is approximately three times finer. Similarly, Table 6 presents results for the $Re_\tau = 640$ case, where LBM with $\Delta^+ = 1.00$ —a resolution roughly three times coarser than that of IBLBM ($\Delta^+ = 0.31$)—demands nearly six times the CPU time per eddy turnover.

Thus, IBLBM remains a competitive alternative to conventional LBM, particularly when wall-resolution requirements are taken into account, offering a favorable balance between efficiency and accuracy.

3.3 Turbulent Poiseuille Duct Flow

To further investigate the 6th order IBLBM applicability with two inhomogeneous directions, the method is applied to simulate turbulent Poiseuille flow through a square duct at $Re_\tau = 360$. For duct flow, Re_τ is defined based on the duct height H as $Re_\tau = u_\tau H/\nu$.

The computational domain of the conduit is defined as $2\pi H \times H \times H$ in the x , y , and z directions. No-slip boundary conditions are applied at all four duct walls, while a periodic boundary condition is imposed in the streamwise (x) direction. A nonuniform grid, generated using the same hyperbolic tangent function as in the turbulent Poiseuille channel flow, is employed in the wall-normal directions (i.e., the y - and z -axes). Sixth-order interpolation is used for both laminar and turbulent duct flow simulations.

Previous studies by Kang and Hassan (2013) and Lee et al. (2018), both using a uniform grid, indicated that the D3Q19 lattice model generates additional corner vortices in turbulent duct flows. Furthermore, Lee et al. (2018) demonstrated that even in the laminar regime, the D3Q19 model produces an artificial secondary flow pattern.

Figure 12 illustrates the predicted streamwise velocity contours and secondary flow patterns for laminar flow at $Re_\tau = 30$ using the D3Q19 and D3Q27 lattice models. The results reveal that the D3Q19 model introduces spurious secondary flow, whereas the D3Q27 model, as expected, remains free of such effects. The magnitude of the secondary flow predicted by the D3Q19 model is on the order of 10^{-4} relative to the streamwise velocity. Despite employing a nonuniform grid (64×64), the findings align well with those of Lee et al. (2018). Consequently, the D3Q27 model is chosen for simulating turbulent duct flow.

Table 7 presents the grid configuration and lattice size employed for the turbulent duct flow simulation. In the streamwise direction, the grid spacing is set to $\Delta^+ \sim 5$, while Δ^+ in the wall-normal direction, is maintained within the range of 0.35 to 5.03 to ensure consistency with direct numerical simulation results.

Figure 13 (a) and (b) depict the instantaneous and mean velocity contours, along with the secondary flow vectors, for turbulent duct flow at $Re_\tau = 360$. The impact of secondary flow

Table 5 Simulated number of eddy turn over time (nETT) of $Re_\tau = 180$ channel flow on four NVIDIA V100 GPUs for LBM (with grid number $360 \times 112 \times 180$ and $1080 \times 336 \times 540$) and IBLBM (with grid number $192 \times 192 \times 192$) in different order interpolations. (nETT = $N_{step} \Delta t / (h/u_\tau)$), $N_{step} = 10^*$; *; estimated based on the grid size)

$Re_\tau = 180$ grid size resolution	LBM 7.26×10^6 $\Delta_{x,y,z}^+ = 3.2$	LBM 1.96×10^8 $\Delta_{x,y,z}^+ = 1.07$	IBLBM 7.07×10^6 $\Delta_{x,y,z}^+ = 6, 0.30 \sim 3.83, 3$		
	2nd-order		4th-order	6th-order	
CPU time (s)	20.3	548.1*	21.4	32	37.8
nETT	0.555	0.185*	0.0521	0.0521	0.0521
CPU time(s)/nETT	36.5	2960*	411	615	726

Table 6 Simulated number of eddy turn over time (nETT) of $Re_\tau = 640$ channel flow on four NVIDIA V100 GPUs for LBM (with grid number $1088 \times 340 \times 544$ and $4009 \times 1292 \times 2068$) and IB-LBM (with grid number $832 \times 512 \times 464$) in different order interpolations. (nETT = $N_{step} \Delta t / (h/u_\tau)$), $N_{step} = 10^4$, *: estimated based on the grid size)

$Re_\tau = 640$ grid size resolution	LBM		IB-LBM	
	2.01×10^8 $\Delta_{x,y,z}^+ = 3.8$	1.1×10^{10} $\Delta_{x,y,z}^+ = 1.$	1.98×10^8 $\Delta^+ = 4.92, 0.31 \sim 5.47, 4.41$	
time (s)	209	11,468*	296	452
nETT	0.157	0.0413*	0.0128	0.0128
CPU time(s)/nETT	1,331	377,485*	23,103	35,279
				58,773

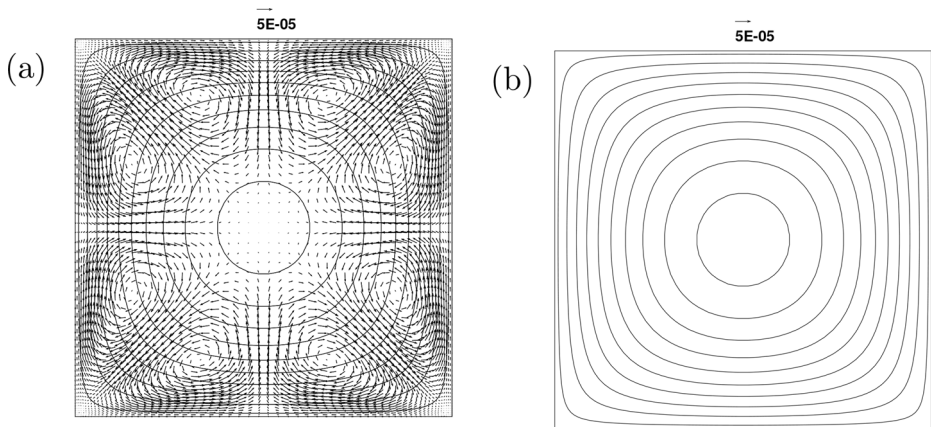


Fig. 12 The predicted streamwise velocity contours and the secondary flow at $Re_\tau = 30$ duct flow with 6th-order IBLBM. (a) D3Q19 lattice model (b) D3Q27 lattice model

on the mean streamwise velocity is most pronounced in the bulging of velocity contours toward the corners, as observed in Fig. 13 (b). This phenomenon occurs as the secondary flow redistributes high-momentum fluid from the central region to the corners.

The predicted mean streamwise velocity and turbulence intensities along the wall bisector are presented in Figs. 14 and 15 and compared with the direct numerical simulation (DNS) data of Moser et al. (1999) for channel flow. The results exhibit excellent agreement with the benchmark solutions, demonstrating the accuracy of the present approach.

The detailed distributions of the predicted streamwise velocity and secondary flow velocity in various span cross sections are presented in Figs. 16 and 17. These results are compared with duct flow data at $Re_\tau = 300$ from Gavrilakis (1992) and Pattison et al. (2009).

The predicted wall shear stress is shown in Fig. 18 and is compared with the results of Gavrilakis (1992) and Hsu et al. (2012). Overall, the predicted results agree well with the benchmark solutions despite slight differences in Reynolds numbers.

4 Conclusion

This study presents numerical simulations of turbulent channel and duct flows using the interpolation-based Lattice Boltzmann Method (IBLBM) with nonuniform meshes. The investigation begins with laminar Poiseuille flow featuring wall injection and suction, where analytical solutions are used as a benchmark to assess the effect of interpolation order. In this configuration, the maximum velocity shifts toward the suction boundary, producing a more pronounced near-wall velocity gradient that resembles turbulent boundary layers—conditions under which nonuniform meshes offer distinct advantages. Results show that IBLBM with fourth-order interpolation and higher achieves accuracy comparable to conventional uniform-mesh LBM while requiring fewer grid points, whereas second-order interpolation yields reduced predictive accuracy.

For turbulent channel flows up to $Re_\tau = 1000$, nonuniform grids were employed with $\Delta x \sim 5$ in the periodic directions and $0.3 \lesssim \Delta x \lesssim 5$ in the wall-normal direction. Pre-

Table 7 Grid setting for the turbulent duct flow at $Re_\tau = 360$

Re_τ	Domain size	Grid number	Δx^+	$\Delta y^+(\Delta z^+)$	$y_{iw}^+(z_w^+)$	lattice size δ^+
360	$2\pi H \times H \times H$	$448 \times 128 \times 128$	5.05	$0.7 \sim 5.03$	0.35	0.7

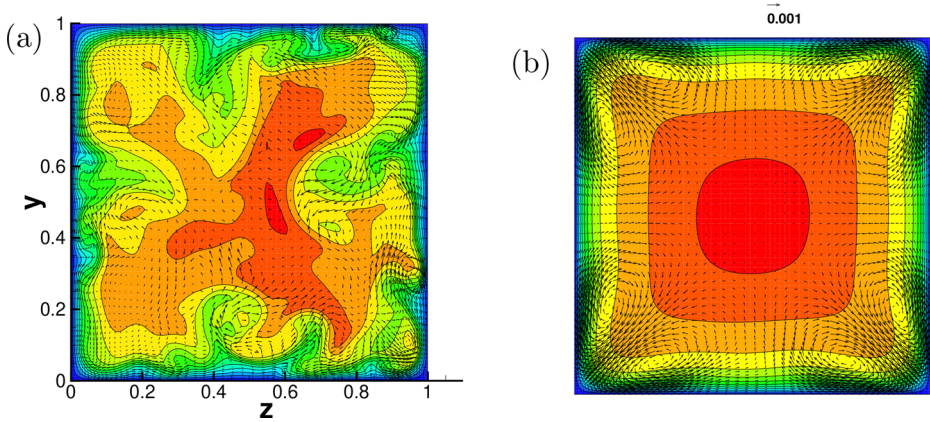


Fig. 13 The streamwise velocity contours and the secondary flow at $Re_\tau = 360$ duct flow with 6th-order IBLBM. (a) Instantaneous (b) mean

Fig. 14 The mean streamwise velocity profile along the wall bisector at $Re_\tau = 360$ duct flow

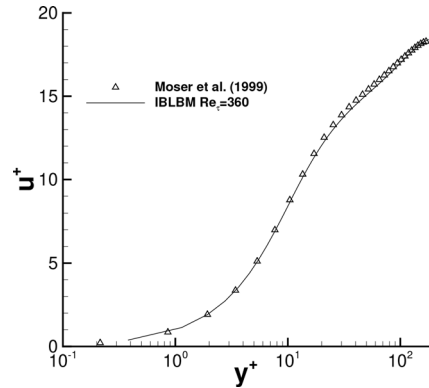
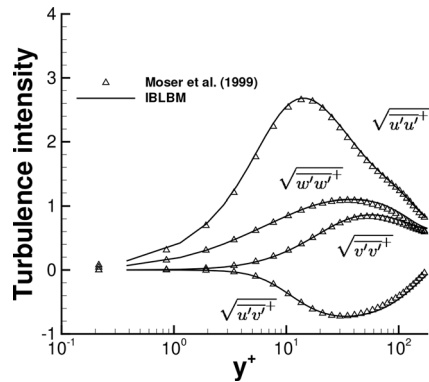


Fig. 15 Distribution of turbulent intensity and the primary Reynolds stress at $Re_\tau = 360$ duct flow



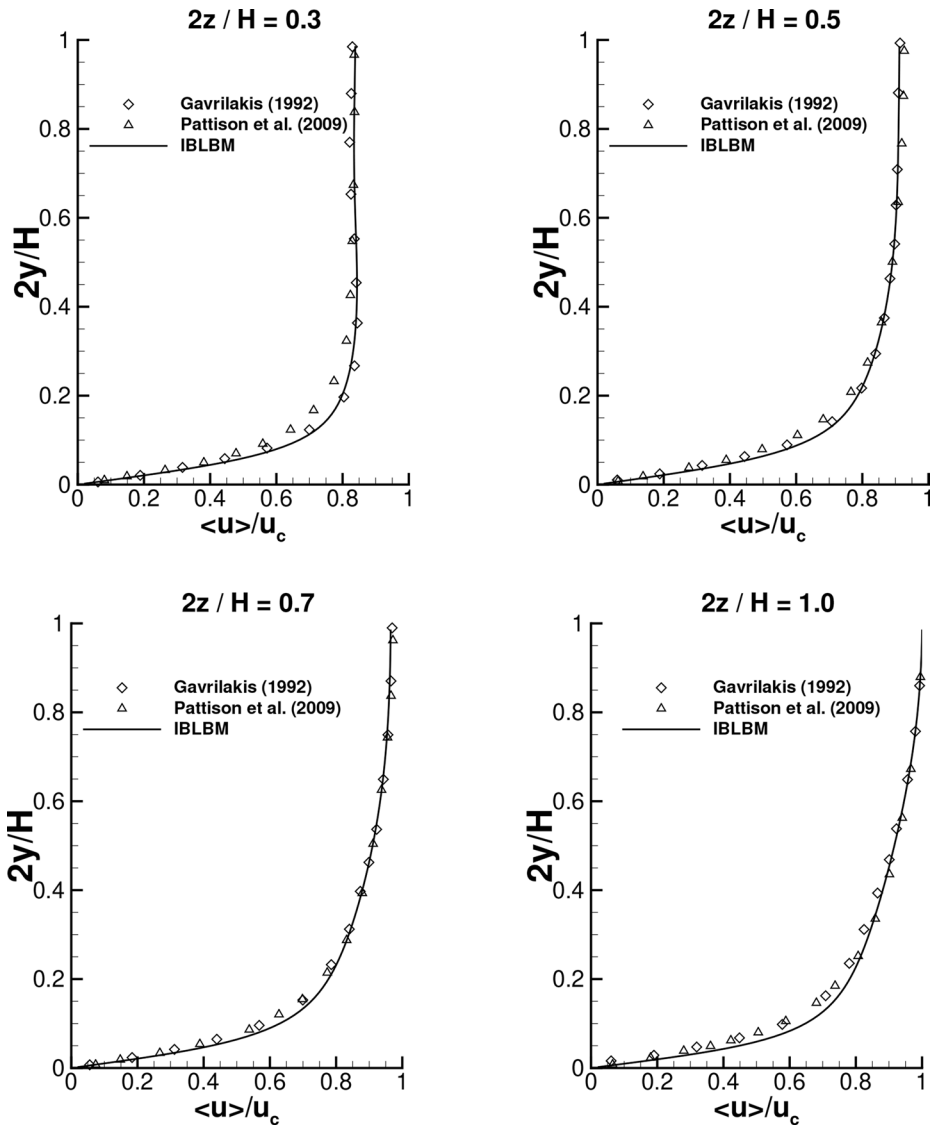


Fig. 16 The streamwise velocity profiles at different locations at $Re_\tau = 360$ duct flow

dicted mean flow and turbulence statistics were compared with benchmark DNS data. At $Re_\tau = 180$, tests with second-, fourth-, and sixth-order interpolation showed that second-order interpolation caused overprediction of streamwise velocity and incorrect turbulent vorticity distribution. At higher Reynolds numbers ($Re_\tau = 640$ and 1000), sixth- and eighth-order interpolations produced comparable results; hence, the sixth-order scheme was adopted for subsequent simulations. The predictions closely matched benchmark DNS, accurately resolving the viscous sublayer and reproducing key turbulence statistics, including mean velocity profiles, Reynolds stresses, and turbulent kinetic energy budgets.

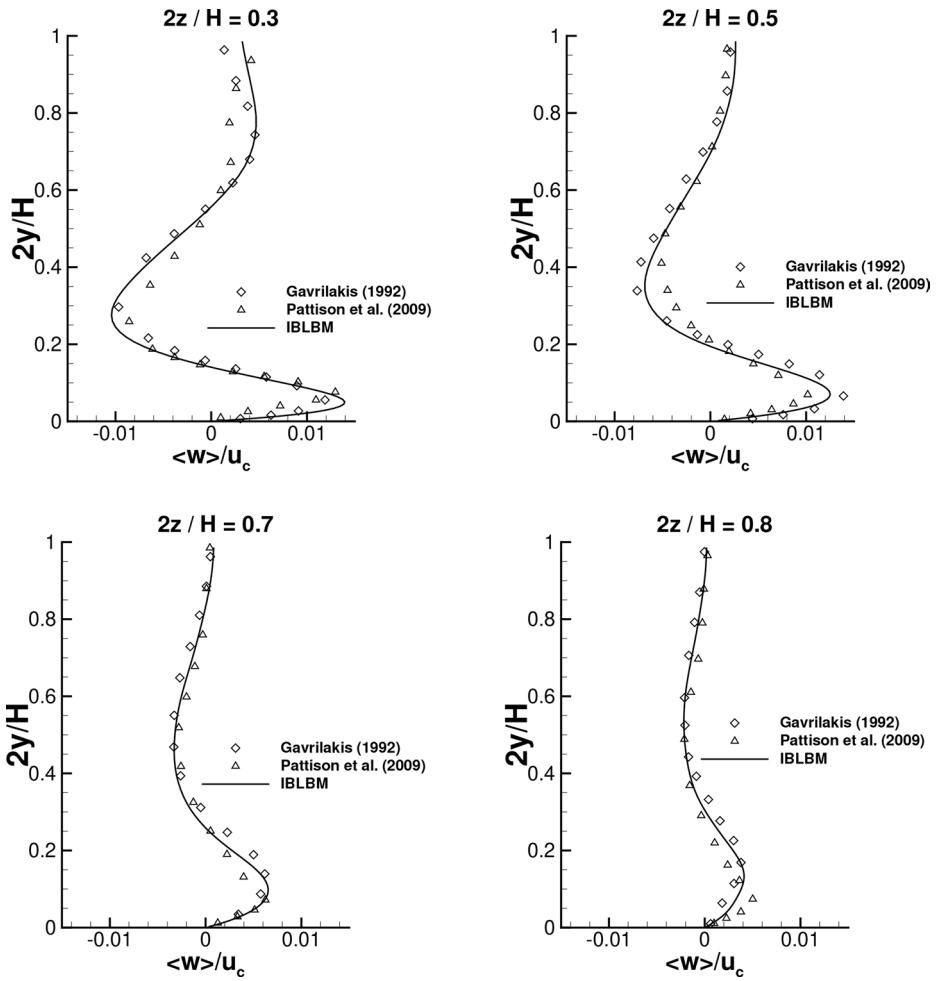
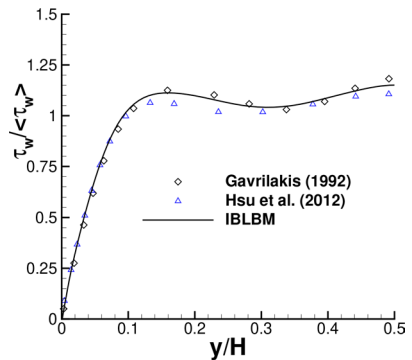


Fig. 17 The spanwise velocity profiles at different locations at $Re_\tau = 360$ duct flow

Fig. 18 Wall shear stress distributions



The method was further applied to turbulent duct flow with two inhomogeneous directions, confirming the superiority of the D3Q27 lattice model over the D3Q19 model in avoiding spurious secondary flows. Predicted mean velocity contours and secondary flow patterns agreed well with reference data, validating the approach for more complex geometries.

Performance analysis shows that, for the same total grid count, the computational overhead of IBLBM remains of the same order across interpolation schemes. For the 7- and 200-million-grid cases at $Re_\tau = 180$ and 640, LBM with $\Delta^+ \sim 3$ is approximately 1.9 and 3.7 times faster, respectively, than sixth-order IBLBM. However, to match the wall resolution of IBLBM ($\Delta^+ \sim 0.3$), adopting $\Delta^+ \sim 1$, LBM becomes four to six times slower, substantially increasing computational cost and negating its performance advantage.

Overall, the results demonstrate that IBLBM with nonuniform meshes is a robust and accurate approach for simulating wall-bounded turbulent flows. By enabling high near-wall resolution with fewer grid points, it achieves an optimal balance between computational efficiency and accuracy, making it particularly effective for high-Reynolds-number simulations where wall resolution is critical.

Author Contributions B.-X., S.-W. and Y.-H., and C.-A. conducted the simulations. B.-X., Y.-H., and C.-A. wrote the main manuscript text, and B.-X., S.-W., Y.-H., and C.-A. prepared figures. All authors reviewed the manuscript. C.-A. secured the funding.

Funding Open access funding provided by National Tsing Hua University. Bo-Xiao Jin is supported by the National Tsing Hua University Graduate Student Scholarship, Hsinchu, Taiwan. The Taiwan National Science and Technology Council sponsors the project under grant number NSTC 113-2221-E-007-129.

Data Availability The authors declare that the data supporting the findings of this study are available from the corresponding authors on reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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